

BloodIQ: Research Project Plan & Team Division

Based on the Hybrid GRU-LightGBM Prescriptive Digital Twin methodology using real MIMIC-IV data, the project is divided into four highly specialized roles. This ensures parallel development and allows each member to contribute a specific technical section to the IEEE research paper.

Member 1: Data Engineering & Pipeline Architect (The Data Lead)

Goal: Replace all synthetic data by building a robust ETL pipeline for real clinical data.

Key Responsibilities:

- **MIMIC-IV Extraction:** Write SQL/Pandas scripts to query the `inputevents` table in MIMIC-IV. Filter specifically for Packed Red Blood Cells (PRBC) and Platelet transfusions.
- **Time-Series Aggregation:** Group patient-level transfusion events into daily aggregated demand signals.
- **Feature Engineering:** Inject exogenous variables that the hybrid model needs (day of week, holidays, seasonal indicators).
- **Paper Contribution:** Write the 'Dataset Strategy and Preprocessing' section of the paper.

Member 2: Predictive Modeling (The ML Lead)

Goal: Build the State-of-the-Art Hybrid forecasting model.

Key Responsibilities:

- **Model Design:** Implement the Hybrid GRU + LightGBM architecture using PyTorch and LightGBM.
- **Training & Tuning:** Train the model on the output of Member 1's pipeline. Handle hyperparameter tuning.
- **Benchmarking:** Train baseline models (ARIMA, standard LSTM, XGBoost) to prove the Hybrid model is superior.
- **Paper Contribution:** Write the core 'Predictive Methodology' and 'Model Evaluation' sections.

Member 3: Operations Research & Optimization (The Digital Twin Lead)

Goal: Convert the demand forecast into actionable business decisions (Prescriptive AI).

Key Responsibilities:

- **Digital Twin Simulation:** Build the inventory tracking state machine following biological FEFO rules.
- **Optimization Solver:** Write a linear programming or heuristic solver (SciPy/PuLP) to calculate optimal daily transfers.
- **Risk Calculation:** Formulate mathematical logic for Wastage Risk and Shortage Risk.
- **Paper Contribution:** Write the 'Prescriptive Optimization and Digital Twin' section.

Member 4: Full-Stack Integration & Dashboarding (The UI/UX Lead)

Goal: Bring the math to life via an interactive application and generate visual figures for the paper.

Key Responsibilities:

- **API Development:** Wrap ML and Digital Twin logic in FastAPI endpoints.
- **Dashboard Creation:** Build the frontend to visualize live inventory, demand curves, and Prescriptive Alerts.
- **Data Visualization:** Generate publication-ready graphs (Plotly/Matplotlib) for the paper.
- **Paper Contribution:** Write the 'System Architecture' section and handle all major figures/UI screenshots.